Primary 3 · Science

Science · Set 2 — Answer Key

80 marks · 1 h 15 min · Focus practice

For parents. Give this to the pupil only after the paper is finished. Written answers: accept any answer with the same meaning, in a complete sentence with correct tense.

Section A: Multiple-Choice Questions (40 marks)

No.	Answer	Marks
1	(2) Chair	2
2	(3) Needing air, food and water	2
3	(3) Root	2
4	(3) Fish	2
5	(3) Cat	2
6	(2) They can make their own food using sunlight	2
7	(3) Plastic	2
8	(2) Glass	2
9	(2) A metal spoon	2
10	(3) Seedling	2
11	(3) Mosquito	2
12	(2) Tadpole	2
13	(1) Which car design rolls the farthest	2
14	(1) Make sure that factor does not change between tests	2
15	(1) Plant A	2
16	(3) Wood	2
17	(2) A rock	2
18	(3) Rabbit	2
19	(2) Mushroom	2
20	(3) Mosquito — 4 stages	2

Section B: Structured Questions (16 marks)

No.	Answer	Marks
21(a)	False	1
21(b)	True	1
22(a)	True	1
22(b)	False	1
23	Stagnant water (e.g. in a flower pot tray or an open container)	2
24	Waterproof (or flexible / has good grip / strong / durable / hard-wearing)	2
25	(1) The amount of soil in each pot	2
26	(2) To check that the results are similar each time (reliable)	2
27	By feeding on dead or decaying matter (not by using sunlight)	2
28(a)	False	1
28(b)	True	1

Section C: Open-Ended Questions (24 marks)

No.	Answer	Marks
29	During the pupa stage, the caterpillar's body changes completely into a butterfly — this is a complete change in body shape. While inside the pupa, it does not eat and it does not lay eggs; it stays still while its body changes. — Marking guide: 1 mark for describing the complete body change inside the pupa, 1 mark for saying it does not feed during this stage, 1 mark for saying it does not lay eggs during this stage. | Hint: Say what changes about its BODY, and whether it eats or lays eggs.	3
30	A cockroach nymph already looks like a small adult cockroach and does not change shape very much as it grows — it has no pupa stage. A mosquito, however, goes through a pupa stage where its body changes completely from a larva into an adult, which the cockroach never goes through. — Marking guide: 1 mark for describing the cockroach nymph as already resembling the adult with no big body change, 1 mark for describing the mosquito's pupa stage as a stage of major body change, 1 mark for a clear comparison between the two. | Hint: Compare BOTH animals: say what happens to one, then what happens to the other.	3

31	A tadpole starts life breathing through gills and swimming with a tail. As it grows into a frog, it grows four legs, loses its tail, and its gills are replaced by lungs for breathing air. A chick already looks similar to a hen and mostly just grows bigger, without losing a tail or changing how it breathes. — Marking guide: 1 mark for describing a specific tadpole body change (e.g. gills to lungs, losing the tail, or growing legs), 1 mark for noting the chick does not go through this kind of change, 1 mark for a clear comparison. | Hint: Compare BOTH animals: say what happens to one, then what happens to the other.	3
32	Mosquito eggs hatch into larvae that live in water, and the larvae become pupae, also in water, before becoming adult mosquitoes. Without stagnant water for the eggs to be laid in, or for the larvae and pupae to develop in, fewer mosquitoes can grow into adults, so there is less chance of mosquitoes spreading dengue fever. — Marking guide: 1 mark for noting that mosquito eggs/larvae/pupae need water to develop, 1 mark for linking removing water to fewer mosquitoes growing into adults, 1 mark for a clear overall explanation connecting this to preventing dengue. | Hint: Think about which stages of the mosquito's life cycle happen in water.	3
33	Although both mushrooms and ferns can reproduce using spores, the key difference is that plants like ferns can make their own food using sunlight, but fungi like mushrooms cannot make their own food — they get their food by feeding on dead or decaying matter instead. — Marking guide: 2 marks for stating the key difference clearly for BOTH sides — fungi like mushrooms cannot make their own food, but plants like ferns can make their own food using sunlight (1 mark for each side). 1 mark for noting that having spores alone does not make them the same kind of living thing (ferns are plants, mushrooms are fungi). | Hint: Spores are not the key difference — think about how each one gets its FOOD.	3

34	Only the type of paper (thin or thick) was changed. The length, width and the weights used were kept the same for every strip. This makes it a fair test because any difference in how much weight a strip holds before tearing is caused only by the type of paper, not by any other factor. — Marking guide: 1 mark for correctly stating what was changed (the type of paper), 1 mark for correctly stating what was kept the same (length, width, weights used), 1 mark for explaining why this makes the test fair. | Hint: Use the frame: Only the ___ was changed, so any difference in ___ is caused by ___. | Explain: Changed: the type of paper (thin or thick). Kept the same: the length, width and weights used. Measured: how much weight each strip held before it tore. Don't mix these up — the weights are part of what was kept the same (every strip was tested the same way), not the thing that was changed.	3
35	Only the material wrapped around the cup was changed. The amount of water, the starting temperature, the cup, and the time waited were all kept the same. This makes it a fair test because any difference in temperature after 10 minutes is caused only by the material used, not by any other factor. — Marking guide: 1 mark for correctly stating what was changed (the material), 1 mark for correctly stating what was kept the same (amount of water, starting temperature, cup, time), 1 mark for explaining why this makes the test fair. | Hint: Use the frame: Only the ___ was changed, so any difference in ___ is caused by ___. | Explain: Changed: the material wrapped around the cup (foam, cloth or plastic). Kept the same: the amount of water, the starting temperature, the cup, and the time waited. Measured: the temperature of the water after 10 minutes. Don't mix these up — temperature is what was MEASURED at the end, not what was changed.	3
36	Canvas fabric is strong, so it does not tear easily even when carrying heavy books. It is also flexible, so it is comfortable to carry and can bend without cracking. These properties make it suitable because a school bag needs to hold weight and be comfortable to use every day. — Marking guide: 1 mark for a correct property (e.g. strong, does not tear), 1 mark for a second correct property with its reason (e.g. flexible, so comfortable or does not crack), 1 mark for a clear overall explanation linking the properties to the purpose. | Hint: Use the frame: ___ is suitable because it is ___, so ___.	3